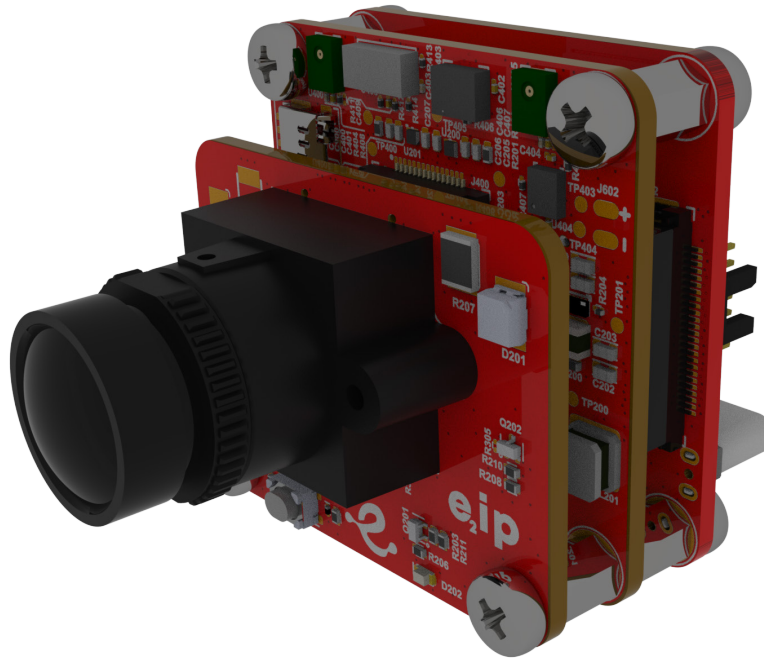


Edge AI Sensing Kit



Explore the next frontier of intelligent Edge AI computing

The Edge AI Sensing Kit is a high-performance, versatile solution designed for edge AI applications leveraging machine vision in its first instance. Featuring the groundbreaking edge AI MCU from ST, this MCU comes with an integrated AI neural network and machine vision system, to enable real-time image analysis, object detection, and event recognition. Its advanced hardware and software integration ensures minimal latency and comes with a compact form factor and energy-efficient design.

Key Features & Benefits



Performant

Dedicated AI functionality for real-time decision-making



Compact & Efficient

Small footprint and energy efficient

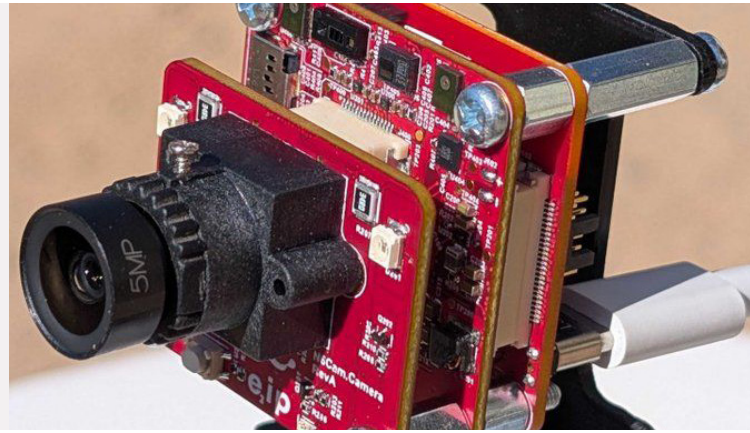


Cost Effective

Cost effective to facilitate edge deployment

AI at the Edge

On-device AI processing with real-time analytics for low-latency.



What's included

STM32N657X0H3Q

STM32N657X0H3Q contains a Neural-Art Accelerator™, an ST-developed Neural Processing Unit.

Image Sensor IMX335-LQN & M12 Lens Mount & Lens

Connectivity Options

USART, USB and WIFI

Software Ecosystem Tools

ST Edge AI Suite

Industries



Medical



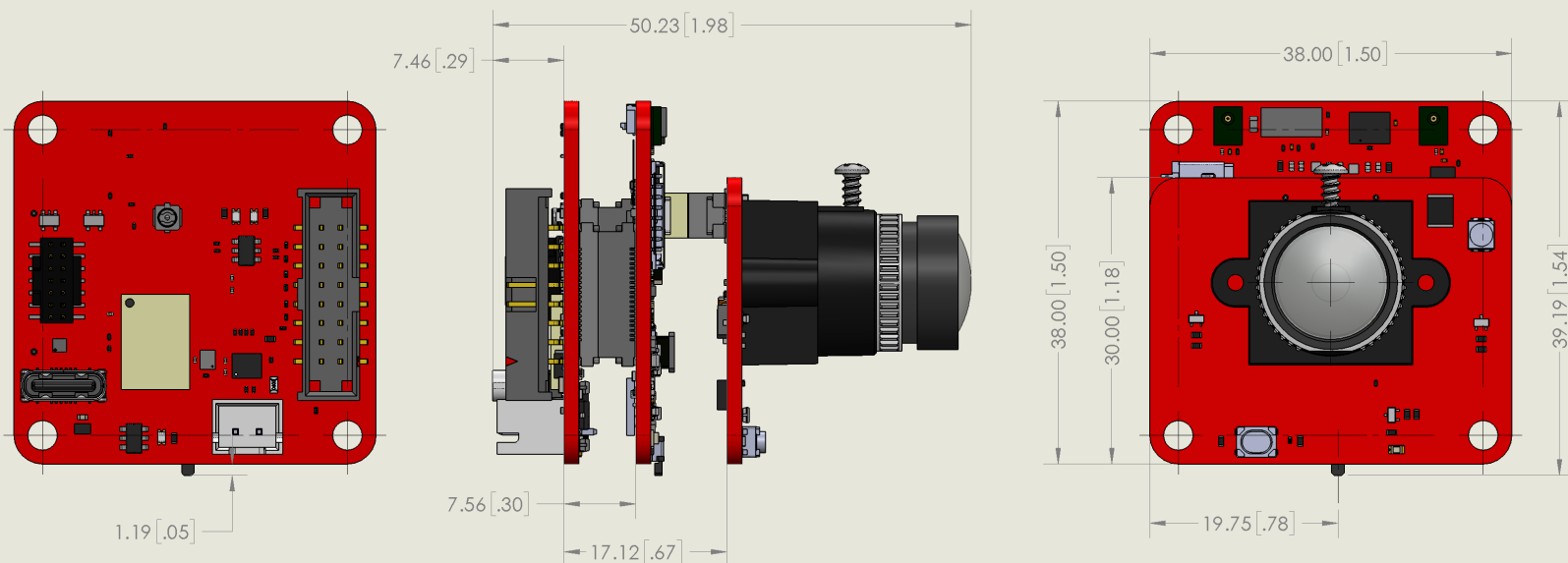
Industrial



Smart Cities



Manufacturing



Accelerating AI innovation at the Edge
for general sensing and machine vision
applications

Specifications

Hardware

Compute Module

MCU

STM32N657

Arm Cortex-M55 @ 800MHz

4.2MB SRAM

Neural-ART accelerator 600 GOPS

Camera ISP + MIPI-CSI2 2-lanes

H-264 encoder: 1080p30, 720p60

HW: JPEG, crypto

External RAM

APS256XXN-OBR-BG

32MB PSRAM

External Flash MX25UM51245GXD100	64MB NOR flash
micro SDCard	-
Sensors	
Human Presence Sensor 1x STHS34PF80	80° FOV
	IR: 5 µm and 20 µm (human)
	up to ~4 meters range
	able to detect stationary objects
Time-of-Flight sensor 1x VL53L7CH	90° Diagonal FOV (60°x60°)
	IR: 940 nm
	up to 350 cm range
	ToF 8x8 zones, up to 30Hz
Inertial Sensor 1x LSM6DSO32	6DOF IMU
	up to: +/-32g, +/-2000dps
	smart FIFO up to 9kB
	advanced functions (motion, tilt)
Microphones 2x MP34DT05-A	MEMS microphones
	spacing = 22mm
	AOP = 122.5 dB SPL
	SNR = 64 dB

IO Module

USB-C

USB-C connector
(back)

Data: USB DRP / HS

Power: 5V @ 1A

Expansion

Header

STMM-108-02-G-D-SM

IO, I2C, SPI, UART, Power

WIFI dual-band

Type 2GF (Murata)

2.4GHz & 5GHz (dual-band)

PCB antenna (or external)

Camera Module

Camera Sensor

Sony IMX335-LQN

5Mp RGB, HDR, rolling shutter

M12 mount + lens

lens: ~90° Horizontal

no IR filter (for night vision)

IR Active Light

2x IR LEDs

For night vision (active)

User LED

1x Red LED

Front facing

User Push-Button

Front facing

Software

ST EDGE AI Suite

Support for multiple object detection models: build, train, and deploy custom models using computer vision models (YOLO, etc).



**Transforming the surfaces we
touch in our everyday lives.**

For more information, speak with
a specialist at e₂ip technologies.

We are always looking forward to
hearing from you!

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